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Leveraging NLP approaches to define and implement text relevance hierarchy framework for business news classification

Muhammad Arslan[a]*,Christophe Cruz[a]

[a]*Laboratoire Interdisciplinaire Carnot de Bourgogne (ICB), 9 Avenue Alain Savary, 21000 Dijon, France*

Abstract

The management of vast amounts of business news data is becoming increasingly important for companies today. To address this challenge, this paper proposes a text relevancy hierarchy framework that enables companies to assess the relevance of business news texts to their specific operations and interests. The framework consists of five levels, ranging from determining the contextual closeness of a text to the company's historical data to identifying new business topics that address the current market situation. The paper highlights the criteria used to assess the importance of texts, such as specific categories, organizations, people, locations, and financial figures involved in the news story. Additionally, the paper explores how leveraging Natural Language Processing (NLP) approaches can assist in the implementation of the text relevancy hierarchy, including entity recognition, topic modeling and similarity analysis. By using this hierarchy framework, companies can prioritize and manage their business news information, focus on the most important and relevant texts, and identify new areas of interest.

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Keywords: text relevancy hierarchy; business news classification; business topics; contextual closeness; entity recognition; natural language processing (NLP); taxonomy; information prioritization; similarity analysis

1. Introduction

In today's business landscape, companies are overwhelmed with vast amounts of data. In particular, business news data is crucial for companies to stay informed about industry trends, emerging technologies, competitors, and other factors that can impact their operations and interests [1]. However, the sheer volume of business news data can

* Corresponding author. Tel.: +33 3 80 39 50 00
E-mail address: muhammad.arslan@u-bourgogne.fr

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be overwhelming and difficult to manage effectively. In addition, traditionally, companies filter or classify business data into relevant and irrelevant categories [2] before conducting analysis (see Fig. 1). Relevant data benchmarks for identifying relevant texts are often constructed using existing company taxonomies (i.e. a list of organized concepts (topics) in the form of a hierarchy) that were created years before and may no longer be up to date [2]. This process can result in the potential loss of relevant information that is not accounted for in the taxonomy.

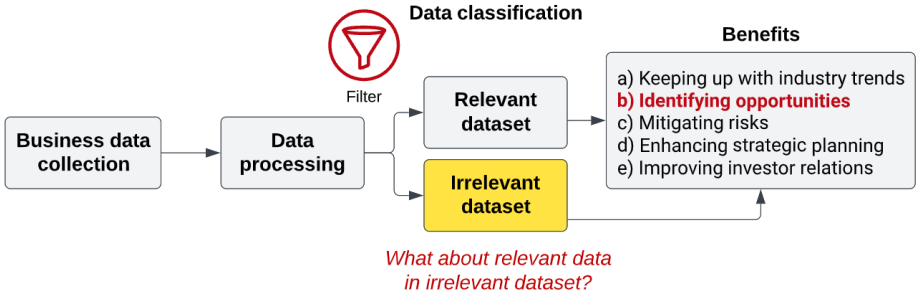


Fig. 1. Deriving value from business data involves different stages of processing

To address this issue, the proposed text relevancy hierarchy framework offers a structured approach to assessing the relevance of business news texts. The framework considers various criteria to determine relevance, such as the contextual closeness of a text to the company's historical data, specific categories, organizations, people, locations, and financial figures involved in the news story, and the identification of new business topics that address the current market situation. The proposed framework consists of five levels that progressively assess the relevance of business news texts. The 1st level involves determining the contextual closeness of a text to the company's historical data. The 2nd level involves assessing whether the text falls within the topics of the company's taxonomy. The 3rd level involves assessing the importance of texts based on specific criteria, such as addressing future business events or involving certain topics, organizations, people, locations, and financial figures. The 4th level involves identifying new business topics from previously classified as irrelevant dataset by the company that address the current market situation and may be relevant to the company, even if they fall outside the current taxonomy. Lastly, at level 5, texts can be considered relevant to each other if they share similar characteristics, such as context (i.e. topic), location, time frame, organization, or authorship.

Furthermore, this paper explores how NLP approaches [3] can assist in the implementation of the text relevancy hierarchy framework. By leveraging the NLP techniques, companies can more efficiently and effectively classify their business news data, focusing on the most important and relevant texts while also identifying new areas of interest. Access to relevant business data offers numerous benefits (see Fig. 1), including staying up-to-date with industry trends, identifying opportunities, mitigating risks, enhancing strategic planning, and improving investor relations [4]. In the context of this research, the focus is on identifying business opportunities using online news data. Eventually, this work provides a structured and systematic approach to business news classification, offering a valuable tool for companies looking to navigate the complex and ever-changing business landscape.

This paper begins with a review of the study's background in Section 2, followed by an introduction of the proposed method in Section 3. Section 4 presents a detailed discussion, and the conclusion is provided in Section 5.

2. Background

In today's digital age, the amount of data available to businesses has increased exponentially [4]. With the rise of the internet and social media, companies have access to vast amounts of business news data from various sources [2]. However, with the sheer volume of information available, it has become increasingly challenging to identify the most relevant news stories and extract meaningful insights from them. To address this challenge, various techniques and frameworks have been proposed in the literature. For example, some studies have focused on using machine learning techniques to classify news articles based on their relevance to a specific topic or industry [5 - 9]. Other studies have proposed using sentiment analysis (i.e. a type of NLP technique) to gauge public opinion and its impact

on business operations [10, 11]. In recent years, NLP approaches have gained popularity as a means to analyze and extract meaningful information from textual data. These approaches have been applied to various areas, including document clustering, entity recognition, and sentiment analysis [5, 10].

Several studies have proposed using NLP techniques to improve the relevance and usefulness of business news data. For example, SENTiVENT [12] is presented to enable supervised information extraction of relevant company-specific events in economic and financial news. It is constructed using the English online business news that were crawled using the business news aggregators Yahoo! Finance API. Another study [13] suggested a method for identifying fake news (i.e. misinformation) by analyzing news titles using a content-based classification approach. To achieve this, the researchers employed a (Bidirectional Encoder Representations from Transformers) BERT model and an LSTM layer. Filtering irrelevant fake news helps business analysts to extract meaningful insights from the relevant news dataset. Moreover, the study [14] presented an intelligent machine learning model based on Multinomial Naive Bayes (MNB) to classify relevant current affairs news stories using the Inverse Document Frequency (IDF) method.

The existing literature does not only contain studies on business news classification based on the textual features captured by IDF, but it also covers the news classification based on the business events identified in news documents. For instance, the study [15] presented a method for extracting valuable information about business events from online news headlines and leads using the clustering, annotation, and classification process. First, they used word embeddings to represent all terms in the corpus as vectors and group them. Second, an expert annotates each group of terms with a corresponding business event. Finally, the information on business events is used as a classifier to identify potentially relevant events from online news data. Another study that is found important for this article is about finding the similarity between a pair of news articles [16]. Goel and Reddy [16] proposed seven different objective similarity metrics such as geography, entities, time, etc. for similarity detection. They used cosine similarity matric and feed-forward neural network for implementation.

The field of text classification has seen numerous techniques and approaches developed over the years, with each proposing different method for categorizing and analyzing textual data. However, despite these advancements, there is a lack of consensus on what constitutes "relevancy" in the context of text classification, leading to confusion and inconsistency in the application of these techniques. Furthermore, while text classification has been widely studied in research settings, its potential applications in business contexts have not been adequately explored. This is a significant gap, given the increasing volume of business news and information available and the growing importance of data-driven decision-making in corporate operations. To address these gaps, the proposed paper puts forward a text relevancy hierarchy framework that provides a systematic and standardized approach to assessing the relevance of business news texts. This framework is designed to enable companies to prioritize and manage information effectively, by defining relevancy in a consistent and meaningful way.

3. Proposed text relevance hierarchy

The text relevancy hierarchy (see Table 1) is a framework used to assess the relevance of business texts to a particular company. The framework consists of five levels which are; 1) Contextual closeness to the company's historical relevant data, 2) Falling within the company's taxonomy topics, 3) Importance based on specific criteria such as addressing future events (i.e. Temporal classification), certain categories (i.e. Hierarchical text classification), organizations, people, locations, and financial figures (i.e. Key-entity classification), 4) Identifying new business topics that relate to the company's operations or interests but fall outside its current taxonomy, and 5) Relevance to other business-relevant texts based on shared characteristics such as topic, location, time frame, organization, or authorship. By using this framework, companies can prioritize and manage information, focus on the most important and relevant business texts, and identify new areas of interest.

To show the implementation of different levels, a dataset by the French company FirstECO containing 108,195 business news texts is used. This dataset was used by the company to identify different business opportunities in news data. This dataset was prepared by crawling the news data from more than 300 different news websites, such as actu.fr, ladepeche.fr, fusacq.com, etc. during the last five years. Later, data pre-processing steps were performed for stop-word removal, filtering of emails, numbers, websites, and special characters using the Natural Language Toolkit (NLTK) Python library [17]. In addition to the company dataset, the taxonomy (i.e. a list of organized topics

that is conventionally used for text classification) of the company contains 2,973 concepts (i.e. abstract ideas) is exploited. Using this dataset and the taxonomy, below, we explain each level of the text relevancy hierarchy.

Table 1. Text relevance hierarchy.

Level	Competency Question	Description
Level 1	What is the relevant business text?	If the text is contextually close to the historical relevant company dataset.
Level 2	How the text is relevant to a company?	If the text comes under the scope of a company`s taxonomy.
Level 3	Which business texts are more or less relevant?	The categorization of business texts at level 3 can be done by utilizing these methods; a) *Temporal classification*: Text addressing future business events is more important and relevant than text addressing past or present events. b) *Hierarchical text classification*: Text on certain topics (concepts of taxonomy) are considered more important (e.g. industrial development, location) than others (e.g. procurement). c) *Key-entity classification*: Certain organizations (companies and corporations), people (CEOs, investors, and government officials), locations, and financial figures (stock prices, revenue, and earnings per share) that are involved in the news story are more important than the rest.
Level 4	Which new business texts, beyond the company's current scope (i.e. taxonomy), could also be relevant to the company?	Texts on new business topics addressing the current market situations can be relevant to the company.
Level 5	Which relevant texts are relevant to a specific business relevant text?	The relevance of a text to another business-related text can be established by identifying their shared characteristics, which could encompass the following: a) Similar context: If two texts cover similar context, they may be considered relevant to each other. b) Similar geographic location: If two texts relate to the same geographic location, they may be considered relevant to each other. c) Similar time frame: If two similar subject matter (i.e. topic) that occurred within a similar time frame, they may be considered relevant to each other. d) Similar organization: If two texts are related to the same organization (i.e. company), they may be considered relevant to each other. e) Similar authorship: If two texts are written by the same author or group of authors, they may be considered relevant to each other.

3.1. Level 1: What is the relevant business text?

When it comes to identifying relevant business text that is similar to a historical relevant company dataset, there are two main approaches: keyword matching and machine learning [3]. Keyword matching involves using techniques like term frequency-inverse document frequency (TF-IDF) to identify common words or phrases between the two datasets. On the other hand, machine learning methods employ supervised or unsupervised learning techniques to identify relationships between the historical relevant company dataset and the new business text [3].

In our case, the keyword-matching technique was used to demonstrate text relevancy detection at level 1. The historical relevant data of the company as described above that was manually classified using Elasticsearch techniques [18] during the last 5 years is used for extracting important business keywords. The KeyBERT model (i.e. an easy-to-use keyword extraction technique that uses BERT embeddings) [19] was used to extract relevant keywords. More details on KeyBERT can be found here [19]. A dictionary of relevant keywords was created (see Fig. 2), and new business texts were identified into relevant or irrelevant categories based on the presence of the relevant keywords.

Fig. 2. Extraction of relevant keywords from a company dataset.

3.2. Level 2: How the text is relevant to a company?

This level focuses on determining the degree of relevancy between a new business text and a company taxonomy, by identifying the number of topics from a company taxonomy that corresponds to the text. For this, the implemented method used cosine similarity (i.e. a commonly used measure for computing the similarity between two texts after their conversion into vectors) [20] to find the similarity (a score between 0 and 1, where 1 indicates the highest similarity) of a text with the concepts of a taxonomy. We have computed the cosine similarity by producing a similarity matrix using the Spacy library [21], and the Sklearn.metrics.pairwise module's cosine_similarity function (http://scikit-learn.org/) between the business text and all the concepts of the taxonomy. Later, the concepts of a taxonomy having the top 5 highest similarity scores with the business text existing in the cosine similarity matrix were retrieved to plot the Fig. 3 below.

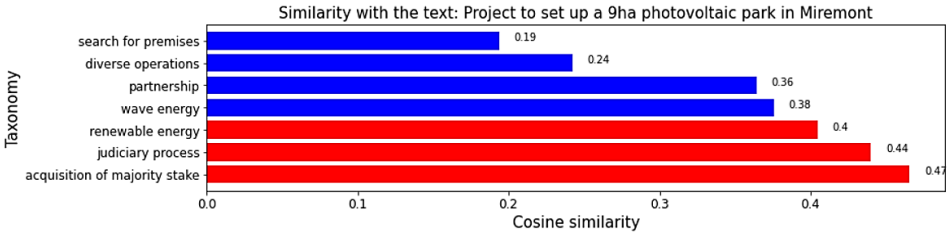


Fig. 3. Checking the relevancy of a business text with different concepts of taxonomy.

3.3. Level 3: Which business texts are more or less relevant?

The categorization of business texts at level 3 can be done by utilizing the methods listed below, which help to distinguish between more or less relevant business texts.

a) Temporal classification: This method involves identifying the past, present, or future events based on the tense of the verbs in the text. The preprocessed text is passed through the Spacy pipeline [21] for extracting verb phrases using noun chunking and identifies the tense of the verbs using their part-of-speech tags. If the tense of any verb in the text is past, present, or future, the text is labeled accordingly (see Fig. 4). However, this approach may not be effective in capturing all variations and nuances in language, and machine learning approaches may be required in such cases to identify events more accurately.

	Text	Event
0	Bayer: 1.6 billion euros was in net profit of 2022 - Tribune.fr	Past
1	Bacardi-Martini deployed a new device called BM Touch - LSA Conso	Past
2	Aveyron: EDF will pay a check for more than 5 million euros to the municipalities in December - actu.fr	Future
3	AXA France is launching a new commercial brand, AXA Next - Les Echos	Future
4	Ramsay Santé continues its development in the Nordic countries - Boursorama	Present

Fig. 4. Temporal classification of business texts

b) Hierarchical text classification: This involves assigning text to specific topics within a taxonomy, where certain topics are considered more important than others. The preprocessed texts are used for the identification of keywords and phrases for determining their topics based on the company's taxonomy. The assigned topics can then be used to prioritize the importance of the news for the business audience (see Fig. 5).

	Text	Category	Level of Importance
0	BASF to invest $10 billion in new Louisiana plant, creating 1,000 jobs	Industrial Development	Very important
1	Japanese company launches new air purifier for improved indoor air quality	Ventilation Device	Important
2	New art exhibit showcases emerging talent from underrepresented communities	Arts and Hobbies	Less important
3	Ford announces plan to shift production to electric vehicles by 2030	Industrial Development	Very important

Fig. 5. Hierarchical text classification based on different categories of taxonomy.

For instance, texts in the "very important" category, addressing different aspects of “Industrial development” can be given more attention, as they are more relevant and impactful than other topics such as “Arts and Hobbies” for business. However, this type of classification is dependent on the company’s interests.

c) Key-entity classification: This type of text classification follows a rule-based approach (i.e. relies on pre-defined rules and heuristics) [22] where articles are labeled based on the presence of specific keywords or entities that are deemed important. For this, the articles are labeled with three levels of importance: very important, important, and less important (see Fig. 6). The criteria for assigning importance levels are based on the presence of certain entities such as the number of vacancies, amount of money, or surface area found in the text. In addition, we can make a priority list of keywords such as “Extension”, “Recruitment”, “Electricity production”, and “Construction”, etc. which are important for business. These keywords can be searched in the texts and their priority can be assigned accordingly.

	Text	Cause	Level of Importance
0	Capgemini has more than 800 vacancies in Lyon and Grenoble in 2023	Number of vacancies found	Very important
1	Extension project of a commercial complex in Beynost	Keyword found	Important
2	€2.5 million to modernize the Rémy-Huckel sports complex in Brumath	Amount found	Very important
3	EDF - CNPE de Penly is authorized to build 1,834m² of premises in Petit-Caux	Surface area found	Very important
4	A bakery in a Paris suburb introduced a new line of gluten-free croissants	Not in the prioroty list	Less important

Fig. 6. Key-entity classification.

3.4. Level 4: Which new business texts, beyond the company's current scope (i.e. taxonomy), could also be relevant to the company?

There exist several methods in the literature for extracting topics from business texts that are not part of a pre-defined taxonomy. These methods include text clustering, topic modeling, and keyword extraction [23, 24]. Text clustering uses unsupervised machine learning algorithms to group similar texts, while topic modeling identifies latent topics in texts. Keyword extraction involves identifying the most frequently occurring words or phrases. However, it is important to note that these approaches may not always produce accurate and comprehensive topic labels, so it may be necessary to combine multiple approaches or manually review the results to ensure their quality. In our case, to demonstrate the proof-of-concept work of text relevance identification at level 4, we have used a state-of-the-art BERTopic model. BERTopic [23] is a topic modeling technique that utilizes pre-trained BERT language models to generate embeddings for documents. It then applies a hierarchical clustering algorithm to group similar documents into topics. Using BERTopic, we have identified new topics (like a set of different keywords) in the dataset that has been categorized as irrelevant data by the company that can be relevant for business (see Fig. 7).

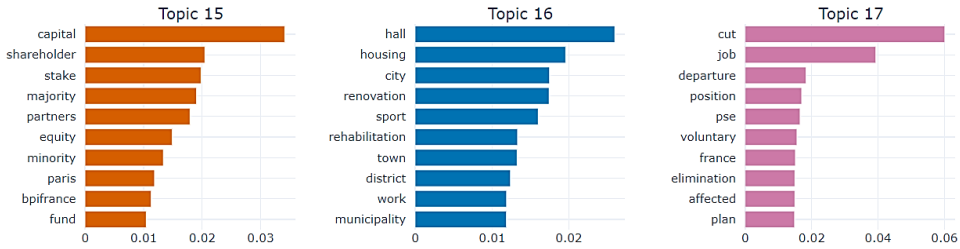


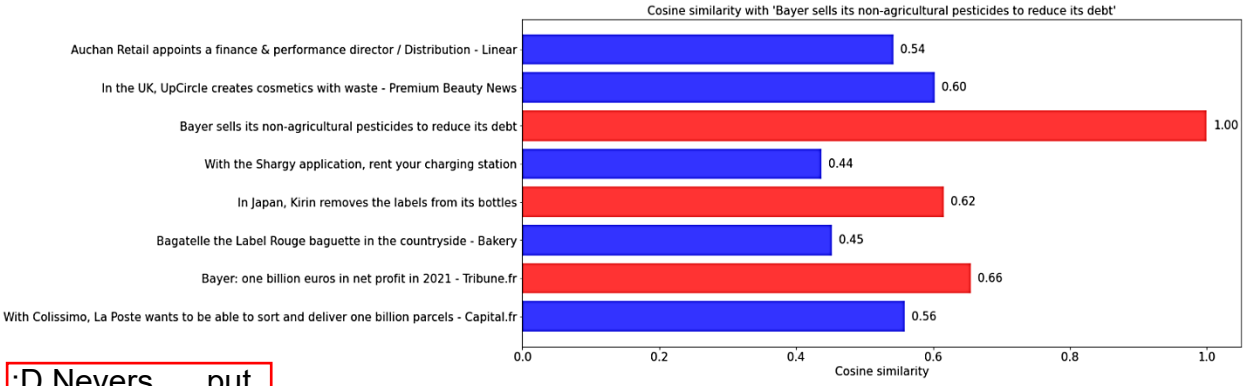
Fig. 7. Identifying new business topics from a dataset.

3.5. Level 5: Which relevant texts are relevant to a specific business relevant text?

To establish the relevance of a business-related text to another, shared characteristics can be identified including: a) similar context (i.e. topic), b) geographic location, c) time frame, d) organization, and e) authorship. If two texts

cover similar context, relate to the same geographic location, occurred within a similar time frame, are related to the same organization, or are written by the same author or group of authors, they may be considered relevant to each other.

a) Similar context: To identify similar texts, we have used a pre-trained model for BERT called "bert-base-multilingual-cased" [25]. Then, we extract the document embeddings of texts using the BERT model. These document embeddings are used to compare the similarity between different business texts (see Fig. 8).



:D Nevers .... put Dijon instead :D

Fig. 8. Identifying the similarity of a relevant text with other relevant texts.

b) Similar geographic location: Assessing the similarity of geographic locations in business texts involves calculating the distances between mentioned locations (see Fig. 9). Texts referring to similar or nearby geographical territories can be deemed important for business.

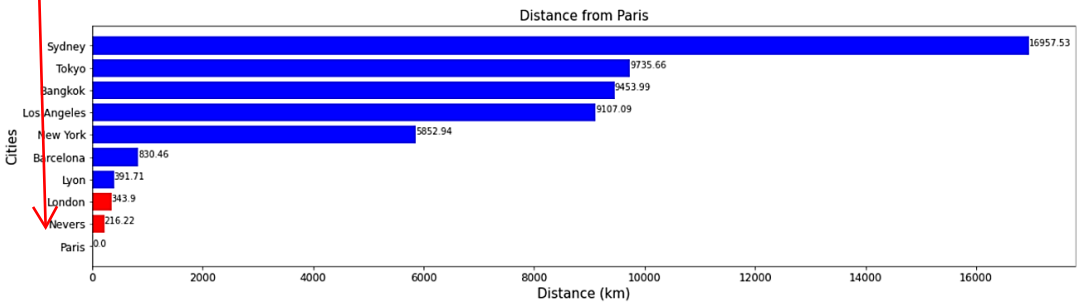


Fig. 9. Identifying the relevancy of a location with respect to other locations.

c) Similar time frame: Texts concerning similar topics occurring within a close time frame can be measured for relevance using cosine similarity, which generates a score between 0 and 1. The score can be visualized in a Fig. 10 to identify highly similar business topics, providing insights for analysis such as trend analysis or recommendation.

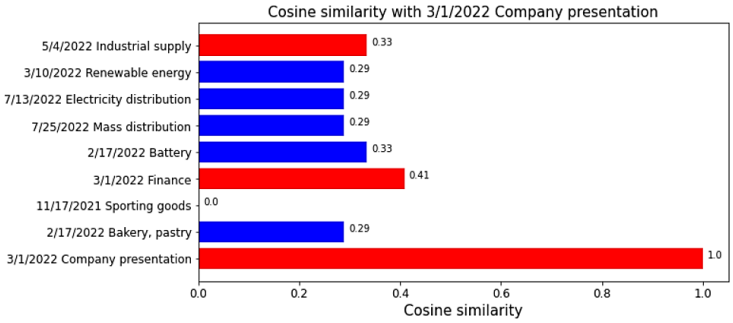


Fig. 10. Identifying similarities between topics and their respective dates and comparing them with other topics and dates.

d) Similar organization: Business texts mentioning the same organization (i.e. company name) may be relevant to each other. We have identified such related texts (see Fig. 11) using techniques like Levenshtein distance [26], which measures the similarity between two texts based on their constituent words or characters (i.e. single-character edits (insertions, deletions, or substitutions) required to transform one string into another).

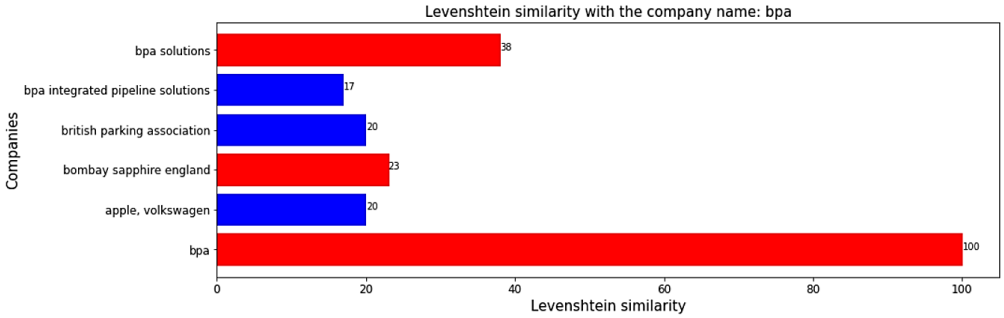


Fig. 11. Identifying the similarity of a company name with the other company names.

e) Similar authorship: When analyzing a corpus of business texts, one important factor to consider is the authorship of each text. If two texts are written by the same author or group of authors, they may be considered relevant to each other, since they likely share similar topics. This is known as "similar authorship." One method of determining whether two texts have similar authorship is to use a string comparison algorithm like the Levenshtein distance [26]. By comparing the Levenshtein distance between two names of authors, we can get a sense of how similar they are in terms of their wording and structure (see Fig. 12).

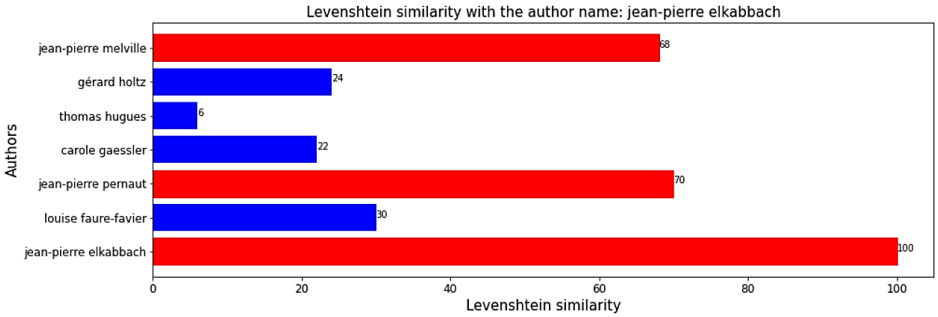


Fig. 12. Identifying the similarity between author names

4. Discussion

The text relevancy hierarchy framework presented in this study offers a useful approach for assessing the relevance of business texts to a particular company. In our case, that is for the French company FirstECO. The framework is based on five levels, each of which has its own criteria for determining the level of relevancy of a text to a company. The 1st level, contextual closeness to the company's historical relevant data, emphasizes the importance of texts that are closely related to a company's historical data. Such texts are likely to be highly relevant to the company's current operations and plans, as they provide valuable insights about the company's interests regarding their business. The 2nd level, falling within the company's taxonomy categories, emphasizes the importance of texts that are classified within the company's taxonomy. This means that the text should be classified under one of the company's existing concepts or topics, and should be labeled accordingly. This level ensures that the company can manage and categorize its information effectively. The 3rd level, importance based on specific criteria such as addressing future events, certain categories, organizations, people, locations, and financial figures, emphasizes the importance of texts that address specific criteria that are relevant to the company. The 4th level, identifying new business topics that relate to the company's interests but fall outside its current taxonomy, emphasizes the importance of identifying new areas of opportunity that are not currently covered by the company's taxonomy. This level ensures that the company can stay up-to-date with emerging trends and topics. The 5th level,

relevance to other business-relevant texts based on shared characteristics such as context, location, time frame, organization, or authorship, emphasizes the importance of texts that share common characteristics with other relevant texts. This level ensures that the company can identify related texts that may provide additional insights.

The purpose of using different levels of text relevancy is to improve the accuracy of classifying relevant information for business purposes. However, relying solely on a single method of text relevancy may not guarantee the complete identification of all relevant texts. For instance, the data classification at level 2 revealed many business texts that contained the names of organizations but were not identified as relevant using the employed approach for text relevance. An example is shown in Fig. 13. To address this issue, we incorporated level 3 of text relevancy, which involves key-entity classification. By identifying the company name and analysing the description of the company's name that exists in the company's database (see Fig. 14), we can extract the specific business topic corresponding to a particular text (e.g. "The Body Shop" in text mentioned in Fig. 13 relates to "Cosmetics"). This process ensures that all elements of the text are processed accurately for proper classification.

	A	B	C	D
1	**No.**	**Text**	**Identified Topic**	**Missed Topic**
2	1	UpCircle is a UK-based company that creates products using waste materials. The company collects coffee grounds from coffee shops in London to create its products, along with other ingredients such as chamomile infusion and olive kernel powder. The company faces growing competition from other businesses, including The Body Shop, which also uses waste materials in its products.	"Waste collection"	"Cosmetics"

Fig. 13. Classification method failed to accurately classify the text with regards to the mentioned company name i.e. "The Body Shop"

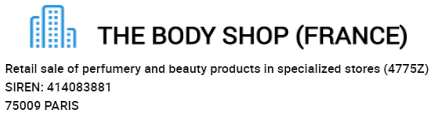


Fig. 14. The company's database contains a mapping of company names to their business descriptions, which were obtained from https://www.societe.com/.

5. Conclusion

As the amount of business news data continues to grow, managing and analyzing this information has become increasingly important for companies. This paper proposes a text relevancy hierarchy framework that enables businesses to assess the relevance of business news texts to their specific operations and interests. The framework provides a systematic approach for evaluating the importance of texts based on contextual closeness to the company's historical data, identifying new business topics, and assessing the involvement of specific categories, organizations, people, locations, and financial figures in the news story. Additionally, the paper explores the application of NLP techniques, to assist in the implementation of the framework. By using this hierarchy framework, companies can prioritize and manage their business news information, focus on the most important and relevant texts, and identify new areas of opportunity. The implementation of the framework using FirstECO's dataset and taxonomy demonstrates its practicality and usefulness in real-world settings.

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